

Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur
FOUR YEAR B. TECH. COURSE
(Revised Curriculum as per AICTE Model Curriculum)
Scheme for Computer Engineering

Sixth Semester:-

S. N.	Subject Code	Subject	Teaching Scheme			Evaluation Scheme			Credits	Category
			L	T	P	CA	UE	Total		
1	BTCME601T	Design and Analysis of Algorithms	3	-	-	30	70	100	3	PCC-CS
2	BTCME601P	Design and Analysis of Algorithms- Lab	-	-	2	25	25	50	1	PCC-CS
3	BTCME602T	TCP/IP	3	-	-	30	70	100	3	PCC-CS
4	BTCME602P	TCP/IP -Lab	-	-	2	25	25	50	1	PCC-CS
5	BTCME603T	Elective-II	3	-	-	30	70	100	3	PEC-CS
6	BTCME604T	Elective-III	3	-	-	30	70	100	3	PEC-CS
7	BTCME605T	Open Elective-I	3	-	-	30	70	100	3	OEC-CS
8	BTCME606P	Emerging Technology Lab -II	-	-	2	25	25	50	1	PCC-CS
9	BTCME607P	Mini Project & Internship	-	-	6	25	25	50	3	PROJ-CS
10	BTCME608	Seminar / Presentations on Current trends	-	-	2	50	-	50	Audit	MC
Total			15	00	14	300	450	750	21	

Elective-II: -

BTCME603.1T - Human Computer Interaction

BTCME603.2T - Internet of Things

BTCME603.3T - Soft Computing

Elective-III: -

BTCME604.1T - Distributed Computing

BTCME604.2T - Machine Learning

BTCME604.3T - Data Warehousing & Mining

Open Electives -I:

BTCME605.1T - Digital Image Processing

BTCME605.2T - Mobile Computing

(Dr. A. N. Thakare)

(Prof. V. P. Thakare)

(Prof. S. J. Dute)

Dr. S. V. Sonelkar

Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur
FOUR YEAR B. TECH. COURSE
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Scheme for Computer Engineering

Sixth Semester:-

Subject : Design and Analysis of Algorithms

Subject Code : BTCME601T

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To design the algorithm and analyze it for any real life problem.

Prerequisite(s): Data Structures, Programming Logic

Course Objectives:

1	To understand the importance of algorithm
2	To analyze the complexity of an algorithm in terms of time and space complexities
3	To understand various problem solving techniques
4	To learn about amortized analysis of algorithms
5	To design and implement various programming paradigms and its complexity

Course Outcomes: After completing the course, students will be able:

CO1	To design and analyse the time and space complexity for any algorithm
CO2	Apply the design techniques of algorithm in solving real life problems
CO3	Apply the design techniques of algorithm using dynamic programming in solving real life problems
CO4	Perform amortize analysis for any algorithm
CO5	Understand NP class of problems and propose approximation algorithms for the same

[08 Hrs]

UNIT I:- Introduction

Algorithms, Examples, Tournament method, Evaluating polynomial functions, pre-processing of coefficients, solving recurrence equations, Asymptotic notations of analysis of algorithms, Complexity

(Signatures)

UNIT II:- Divide & Conquer and Greedy Approaches**[08 Hrs]**

Divide and Conquer method , Strassen's matrix multiplication , Greedy method , Huffman code , Minimum spanning trees , Dijkstra algorithm , Knapsack problem , Job sequencing with deadlines.

UNIT III:- Dynamic Programming Approaches**[08 Hrs]**

Dynamic Programming , Knapsack problem , Matrix Chain Multiplication , longest common subsequence Multistage graphs , All pair's shortest paths , Optimal binary search trees Travelling salesman problem.

UNIT IV:- Amortization**[06 Hrs]**

Randomized Algorithms and Amortized Analysis , Las Vegas and Monte Carlo types , Randomized quick sort and its analysis , Min-Cut algorithm.

UNIT V:- NP Problems**[06 Hrs]**

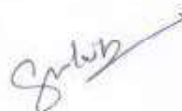
NP, Hard and NP, complete problems , Basic concepts , Reducibility , Vertex cover, 3 , CNF , clique , Hamiltonian cycle , TSP , Approximation algorithms , Vertex cover , TSP.

Text Book:-

1. Thomas H. Cormen, C. Lieserson, R. Rivest, C. Stein, "Introductions to Algorithms", Third Edition, Prentice, Hall/India, 2009.
2. "Fundamentals of Computer Algorithms", Horowitz, Sahani, Rajsekharan, Galgotia Publications.

Reference Books:-

1. M. Tenenbaum, Augestien, "Data Structures using C", Third Edition, Pearson Education, 2007.
2. Sartaj Sahni, "Data Structures, Algorithms and Applications in C++", Universities Press Pvt. Ltd., 2012.



Subject : **Design and Analysis of Algorithms**

Subject Code : **BTCME601P**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
2 Hrs (Practical)	1	25	25	50

Practical List

Practical based on above syllabus.

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Subject: TCP/IP

Subject Code: BTCME602T

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To understand the concept of networking and the different protocols.

Prerequisite(s): Introduction to Computer Network

Course Objectives: (02-04)

1	To understand the network architecture, internetworking concept and internet standards.
2	To know the classless addressing with address resolution protocols.
3	To gain the knowledge of IP layer protocols.
4	Comprehend the TCP and the IP Security.

Course Outcomes: (06)

At the end of this course Student are able to:

CO1	Understand the basics concepts of internetworking.
CO2	Comprehend the various address resolution protocols.
CO3	Use and adapt the IP layer protocols.
CO4	Understand the concept of Transmission control protocol.
CO5	Apply the security to the IP layer.

Unit I: Introduction to Networking

[08 Hrs]

Network architecture-Standards, Comparison of OSI & TCP/IP Model, Connecting Devices, Internetworking concept, Internet Backbones, NAP, ISPs, RFCs and Internet Standards.

Unit II: Internet Addresses & Address Resolution Protocols

[06 Hrs]

Classless Internet address (IPV4), CIDR-Subnetting and Supernetting, IPV6 Addresses: Addressing, Packet Format, Transition from IPV4 to IPV6. ARP, RARP, DHCP.

Unit III: Internet Protocols

[07 Hrs]

IP Datagram-IP Package-IP forwarding and routing algorithms, computing paths, RIP OSPF,BGP ICMPV4, IGMP.

Unit IV: Transmission Control Protocol

[08 Hrs]

TCP header, services, Connection establishment and termination, Interactive data flow, Bulk data flow, Flow control and Retransmission, TCP timers, Urgent Data processing, Congestion control.

Unit V: IP security protocol

[07 Hrs]

IPv6 addresses, Packet format, Multicast, ICMPv6, Interoperation between IPv4 and IPv6-QoS, Auto configuration.

Text Books:

TCP/IP Protocol Suite, Behrouz A. Forouzan, 4th Edition, Tata McGraw Hill Publication.

Reference Books:

Internetworking with TCP/IP - Principles, Protocols, and Architecture, Douglas E. Comer, 5th edition Volume-1, Prentice Hall, 2006.



Subject : TCP/IP

Subject Code : BTCME602P

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
2 Hrs (Practical)	1	25	25	50

Practical List

Practical based on above syllabus.

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Subject : **Human Computer Interaction (Elective-II)**

Subject Code : **BTCME603.1T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To understand the concept of machine learning algorithms & apply to solve the real world problems.

Prerequisite(s): Knowledge of I/O devices, Mobile Application, Web Application,

Course Objectives:

1	Describe what interaction design is and how it relates to human computer interaction and other fields.
2	Use, adapt and extend classic design standards, guidelines, and patterns.
3	Apply core theories, models and methodologies from the field of HCI
4	Types of mobile application along with designing
5	Learn the guidelines in designing user interfaces

Course Outcomes: After completing the course, students will be able to

CO1	Describe the capabilities of both humans and computers
CO2	Design effective dialog for HCI
CO3	Identify the stake holder's requirements and choose the appropriate models
CO4	Develop mobile HCI using mobile elements and tools
CO5	Design Web interfaces using different techniques.

UNIT I :- FOUNDATIONS OF HCI

[08 Hrs]

The Human: I/O channels , Memory , Reasoning and problem solving; The computer: Devices, Memory ,processing and networks; Interaction: Models, frameworks , Ergonomics, styles, elements, interactivity-Paradigms.

UNIT II:- DESIGN & SOFTWARE PROCESS

[09 Hrs]

Interactive Design basics , process , scenarios , navigation , screen design , Iteration and prototyping. HCI in software process , software life cycle , usability engineering , Prototyping in practice , design rationale. Design rules , principles, standards, guidelines, rules. Evaluation Techniques , Universal Design.

Subject : **Internet of Things (Elective-II)**

Subject Code : **BTCME603.2T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To impart the concepts of Internet of Things

Prerequisites: Digital Electronics, Concepts in Computer Engineering

Course Objectives:

1	To provide understanding of design of Computer Architecture and organisation.
2	To provide understanding of issues involved in design of control unit.
3	To develop understanding of concepts of memory organisation and its interfacing.

Course Outcomes

CO1	Students will be able to analyze the development of computers.
CO2	Students will be able to identify the concept of CISC and RISC architecture
CO3	Students to understand Microprocessor, memory and its interfacing

Unit I: Introduction to IoT

[10 hours]

Characteristics of IoT, IOT- ecosystem, Physical design of IoT, Logical design of IoT, Functional blocks of IoT, Communication models/reference model of IOT.

Unit II :

[8 Hours]

Data Link protocols- IEEE 802.15.4e , IEEE 802.11ah,

Transport an session layer protocols – Transport layer TCP, UDP

Session Layer protocols- HTTP, CoAP, XMPP, AMQP, MQTT

Unit III :

[5 Hours]

Domin specific IOT: Home automation, smart cities, Environmental, retail, logistics, Agriculture, Health, Lifestyles

Unit IV : Building blocks of IOT analytics

[6 Hours]

Introduction, IOT Data and BigData, Challenges of IOT analytics applications, cloud based IOT platform, IaaS, PaaS and SaaS paradigms.



UNIT III:- MODELS AND THEORIES

[06 Hrs]

Cognitive models ,Socio-Organizational issues and stake holder requirements ,Communication and collaboration models-Hypertext, Multimedia and WWW.

UNIT IV:- MOBILE HCI

[07 Hrs]

Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools.

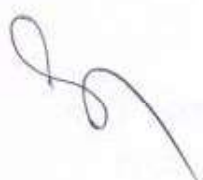
UNIT V:- WEB INTERFACE DESIGN

[06 Hrs]

Designing Web Interfaces , Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow. Case Studies.

Text Books:-

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, "Human Computer Interaction", 3rd Edition, Pearson Education, 2004 (UNIT I , II & III).
2. Brian Fling, "Mobile Design and Development", First Edition , O'Reilly Media Inc., 2009 (UNIT ,IV).
3. Bill Scott and Theresa Neil, "Designing Web Interfaces", First Edition, O'Reilly,2009.(UNIT- V).



Unit V : Tools

[7 Hours]

A review of Tools : Tools and platforms and data reasoning. IOT analytics application and case studies.

Text Books :

1. Internet of Things, Srinivasa K. G, Siddhesh G.M, Hanumantha Raju.R, Cengage Publications
2. Building blocks for IOT Analytics, John Soldatos, River Publishers

Reference Books :

1. Internet of Things, Srinivasa K. G, Siddhesh G.M, Hanumantha Raju. R, Cengage Publications
2. Building blocks for IOT Analytics, John Soldatos, River Publishers







Subject : **Soft Computing (Elective-II)**

Subject Code : **BTCME603.3T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To understand the concept of soft computing and apply to solve the problems.

Prerequisite(s): None

Course Objectives:-

To expose the ideas of Neural networks, fuzzy logic and concepts of Genetic algorithm and its applications to soft computing

Course Outcomes: The students will be able to:

CO1	Recognize the feasibility of applying a soft computing methodology for a particular problem
CO2	Apply fuzzy logic and reasoning to handle uncertainty and solve engineering problems
CO3	Apply Neuro-fuzzy modeling techniques to solve engineering problems
CO4	Apply genetic algorithms to combinatorial optimization problems.
CO5	Apply swarm optimization techniques to solve the particular problems.

Unit1

[08 hrs]

Artificial Neural Networks: Basic concepts - Single layer perception - Multilayer Perception - Supervised and Unsupervised learning – Back propagation networks - Kohonen's self organizing networks - Hopfield network.

Unit2

[08 hrs]

Fuzzy Systems: Fuzzy sets and Fuzzy reasoning - Fuzzy matrices - Fuzzy functions - Decomposition -Fuzzy automata and languages - Fuzzy control methods - Fuzzy decision making.

Unit3

[08 hrs]

Neuro - Fuzzy Modeling: Adaptive networks based Fuzzy interface systems - Classification and Regression Trees -Data clustering algorithms - Rule based structure identification - Neuro-Fuzzy controls -Simulated annealing – Evolutionary computation.

Unit4

[06 hrs]

Genetic Algorithms: Survival of the Fittest - Fitness Computations - Cross over - Mutation - Reproduction -Rank method - Rank space method.

Unit5

[06 hrs]



Swarm Optimization: Ant Colony Optimization, Particle Swarm Optimization, Artificial Bee Colony Optimization

Text Books/Reference Books:-

1. Jang J.S.R., Sun C.T. and Mizutani E, "Neuro-Fuzzy and Soft computing", Prentice Hall 1998.
2. Laurene Fausett, "Fundamentals of Neural Networks", Prentice Hall, 1994.
3. George J. Klir and Bo Yuan, "Fuzzy sets and Fuzzy Logic", Prentice Hall, USA 1995.
4. N. J. Nelsson, "Artificial Intelligence - A New Synthesis", Harcourt Asia Ltd., 1998.
5. D. E. Goldberg, "Genetic Algorithms: Search, Optimization and Machine Learning", Addison Wesley, N.Y, 1989

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8-10-18

Subject: **Distributed Computing (Elective-III)**

Subject Code: **BTCME604.1T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To realize the concepts of Distributed Computing

Prerequisite(s): concept of data structure, networking and operating system

Course Objectives: (02-04)

1	To introduce students to the fundamental problems, concepts, and approaches in the design and analysis of distributed computing systems and applications.
2	To prepare students for an industrial programming environment.

Course Outcomes: (06)

At the end of this course Student are able to:

CO1	To develop and apply knowledge of distributed computing.
CO2	Design a model of distributed computation.
CO3	To understand the concept of logical and global time for message communication
CO4	To analyze the terminology of distributed algorithm
CO5	To apply the knowledge of termination detection algorithm

UNIT I

[7 Hours]

Introduction: Definition, relation to computer system components, Motivation, Relation to parallel multiprocessor / multicomputer system, Message passing versus Shared memory, Primitive for distributed communication , Synchronous versus asynchronous execution.

UNIT II

[7Hours]

Distributed Computation:- A model of Distributed Computation:- A distributed program ,A model of distributed execution, model of communication network , Global state of a distributed system , Cuts of distributed computation , Model of process communication.

UNIT III

[7 Hours]

Logical time:-Introduction, A framework for a system of logical clocks Scalar time, Vector

time, Efficient implementations of vector clock Global state and snapshot recording algorithms:- Introduction ,System model and definitions ,Snapshot algorithms for FIFO channels ,Variations of the Chandy-Lamport algorithm.

UNIT IV

[7 Hours]

Terminology and basic algorithms: - Topology abstraction and overlays, Classifications and basic concepts Complexity measures and metrics, Program structure, Message ordering paradigms:- Asynchronous execution with synchronous communication, Synchronous program order on an asynchronous system

UNIT V

[8 Hours]

Termination detection:-Introduction, System model of a distributed computation, Termination Detection using distributed snapshots, Termination detection by weight throwing, a spanning-tree-based termination detection algorithm , Message-optimal termination detection ,Termination detection in a very general distributed computing model.

Text Books :-

1. Distributed Computing :-- Principles , Algorithm and Systems by Ajay D. Kshemkalyani and Mukesh Singhal.

Reference Books:-

1. Programming Distributed Computing Systems: A Foundational Approach by Carlos A. Varela
Elements of Distributed Computing by Vijay K. Garg



Subject : **Machine Learning (Elective-III)**

Subject Code : **BTCME604.2T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: *To understand the concept of machine learning algorithms & apply to solve the real world problems.*

Prerequisite(s): *Knowledge of Artificial Intelligence*

Course Objectives:

1	To provide a broad survey of different machine learning approaches and techniques
2	To understand the principles and concepts of machine learning
3	To understand neural networks concepts
4	To learn regression and reinforcement learning
5	To develop programming skills that helps to build real world applications based on machine learning

Course Outcomes:

After completing the course, students will be able :

CO1	Solve typical machine learning problems
CO2	Represent data to facilitate learning
CO3	Design and implement various machine learning algorithms for real world applications
CO4	Suggest supervised /unsupervised machine learning approaches for any application
CO5	Handle tools of machine learning

UNIT I: Introduction

[08 Hrs]

Introduction: Machine learning: What and why? Types of Machine Learning, Supervised Learning, Unsupervised Learning, The Curse of dimensionality , Over and under fitting , Model selection, Error analysis and validation , Parametric vs. non, parametric models.

UNIT II: Machine Learning

[07 Hrs]

Types of Machine Learning , Supervised Learning , Classification models , Naïve Bayes Classifier – Decision trees , Support Vector Machines , KNN model , Dimensionality reduction ,PCA.

UNIT III Clustering**[07 Hrs]**

Clustering approaches, Mean Shift clustering , Clustering data points and features – Bi, clustering, Multi, view clustering, K-Means clustering , K-medians clustering , Expectation Maximization (EM).

UNIT IV Neural Networks**[08 Hrs]**

Neural networks , Biological motivation for Neural Network , Neural network Representation , Perceptron – Feed forward networks , Multilayer Networks and Back Propagation Algorithms , Hidden layer representation , Application of neural network.

UNIT V Applications and Tools**[06 Hrs]**

Linear models for regression, Reinforcement Learning, Machine Learning Tools, Engineering applications.

Text Books:-

1. Kevin P. Murphy, "Machine Learning: A Probabilistic Perspective", MIT Press, 2012.
2. Ethem Alpaydin, "Introduction to Machine Learning", Second Edition, Prentice Hall of India, 2010.

Reference Books:-

1. Laurene Fausett, "Fundamentals of Neural Networks, Architectures, Algorithms and Applications", Pearson Education, 2008.
2. Tom Mitchell, "Machine Learning", McGraw, Hill, 1997.
3. C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2007. CSE Dept.
Flexible Curriculum NITTUGCSE21 183
4. Simon Haykin, "Neural Networks and Learning Machines", Pearson 2008



Subject : Data Warehousing & mining (Elective-III) Subject Code : BTCME604.3T

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim : To study the concepts in order to understand how to provide the organization with a reliable source of data and with hidden insights that cannot otherwise be gleaned from large-scale data.

Prerequisites : Database Management Systems

Course Objectives :

1	To understand the various concepts needed to design, develop, and maintain a data warehouse.
2	To analyze data, choose relevant models and algorithms for respective applications.

Course Outcomes

CO1	Understand the need, definition, applications, components, processes & Architecture of a Data Warehouse.
CO2	Learn business requirements, dimensional modeling for designing database schemas for a Data Warehouse.
CO3	Understand Data and Data Mining Principles
CO4	Illustrate frequent pattern mining methods, such as Apriori, ECLAT, and FPGrowth.
CO5	Identify appropriate data mining algorithms to solve real world problems
CO6	Apply different data mining techniques for classification and prediction.

Unit I : [08 hours]

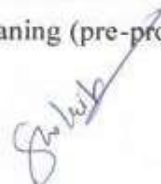
Introduction to data warehousing, evolution of decision support systems, Data warehouse life cycle, building a data warehouse, Data Warehousing Components, Data Warehousing Architecture, Trends in data warehousing Data marts, Metadata

Unit II : [08 Hours]

Online Analytical Processing (OLAP), ROLAP, MOLAP, HOLAP, & DOLAP, OLAP Implementation, OLAP Tools, Multidimensional Databases (MDDBs), SQL Features for DW.

Unit III : [06 Hours]

Introduction to data mining: What is Data Mining? What is the Data Mining Process? Basic Data Mining Tasks, Problem Identification, Data Mining Metrics, Data Cleaning (pre-processing, feature



selection, data reduction, feature encoding, noise and missing values, etc.), Key Issues, Opportunities for Data Mining.

Unit IV :

[06 Hours]

Mining frequent patterns, associations and correlations: Basic concepts, efficient and scalable frequent item set mining algorithms, mining various kinds of association rules – multilevel and multidimensional, association rule mining versus correlation analysis, constraint based association mining.

Unit V :

[08 Hours]

Classification and prediction: Definition, decision tree induction, Bayesian classification, rule based classification, classification by backpropagation and support vector machines, associative classification, lazy learners, prediction, accuracy and error measures.

Text Books:-

1. Han, J. and Kamber, M., "Data Mining - Concepts and Techniques", 3rd Ed., Morgan Kaufmann Series.
2. Ali, A. B. M. S. and Wasimi, S. A., "Data Mining - Methods and Techniques", Cengage Publishers
3. Ponniah Paulraj, "Data Warehousing Fundamentals for IT Professionals", WSE, 2ed., 2010.

Reference Books:-

1. Tan, P.N., Steinbach, M. and Kumar, V., "Introduction to Data Mining", Addison Wesley – Pearson.
2. Pujari, A. K., "Data Mining Techniques", 4th Ed., Sangam Books.

E-Book:- Mining the Social Web Data Mining Facebook Twitter LinkedIn Instagram by Matthew A. Russell & Mikhail Klassen

<https://www.pdfdrive.com/mining-the-social-web-data-mining-facebook-twitter-linkedin-instagram-e185781408.html>

On line TL Material:- <https://nptel.ac.in/courses/106/105/106105174/>









Subject : **Digital Image Processing (Open Elective-I)**

Subject Code : **BTCME605.1T**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To understand the fundamentals of digital image processing and to develop the students ability to apply these tools in various image processing techniques.

Prerequisites : Signals and systems, Calculus and probability, Basic programming skills.

Course Objectives : The student should be made to:

1	Learn digital image fundamentals and will be exposed to simple image processing techniques.
2	Be familiar with image compression and segmentation techniques and learn to represent image in form of feature

Course Outcomes: - Upon successful completion of this course, students will be able to:

CO1	Discuss digital image fundamentals.
CO2	Apply image enhancement and restoration techniques.
CO3	Use image compression and segmentation Techniques.
CO4	Represent features of images.

Unit I:- Introduction

[08 hours]

Fundamental Steps in Digital Image Processing, Components of an Image Processing System, Sampling and Quantization, Representing Digital Images (Data structure), Some Basic Relationships Between Pixels- Neighbours and Connectivity of pixels in image, Applications of Image Processing: Medical imaging, Robot vision, Character recognition, Remote Sensing.

Unit II:-Image Enhancement In The Spatial Domain

[08 Hours]

Some Basic Grey Level Transformations, Histogram Processing, Enhancement Using Arithmetic / Logic Operations, Basics of Spatial Filtering, Smoothing Spatial Filters, Sharpening Spatial Filters, Combining Spatial Enhancement Methods.

Unit III:- Image Enhancement In Frequency Domain

[06 Hours]

Introduction, Fourier Transform, Discrete Fourier Transform (DFT), properties of DFT, Discrete Cosine Transform (DCT), Image filtering in frequency domain.

Unit IV:- Image Segmentation

[06 Hours]

Introduction, Detection of isolated points, line detection, Edge detection, Edge linking, Region based segmentation- Region growing, split and merge technique, local processing, regional processing, Hough transform, Segmentation using Threshold.

Unit V:- Image Compression:

[08 Hours]

Introduction, coding Redundancy, Inter-pixel redundancy, image compression model, Lossy and Lossless compression, Huffman Coding, Arithmetic Coding, LZW coding, Transform Coding, Sub-image size selection, blocking, DCT implementation using FFT, Run length coding.

Text Books:- Rafael C G., Woods R E. and Eddins S L, Digital Image Processing, Prentice Hall, 3rd edition, 2008.

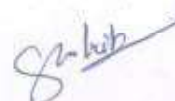
Reference Books:-

1. William K Pratt, "Digital Image Processing", John Willey, 2002.
2. Fundamentals of Digital Image Processing- Anil K. Jain, 2nd Edition, Prentice Hall of India
3. S. Sridhar, Digital Image Processing, Oxford University Press, 2nd Ed, 2016
4. Malay K. Pakhira, "Digital Image Processing and Pattern Recognition", First Edition, PHI Learning Pvt. Ltd., 2011.
5. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, "Digital Image Processing Using MATLAB", Third Edition Tata Mc Graw Hill Pvt. Ltd., 2011.

On line TL Material:-

<http://ceweb.poly.edu/~onur/lectures/lectures.html>.

<http://www.caen.uiowa.edu/~dip/LECTURE/lecture.html>



Subject : Mobile Computing (Open Elective-I)

Subject Code : BTCME605.2T

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
3 Hrs (Theory)	3	30	70	100

Aim: To impart the concepts of Mobile Computing

Prerequisites: Digital Electronics, Concepts in Computer Engineering

Course Objectives:

1	To provide understanding of design of Mobile Wireless Networking
2	To provide understanding of issues involved in Mobile Computing
3	To develop understanding of concepts of Methods involved in Mobile Computing

Course Outcomes

CO1	Students will be able to analyze the Mobile networking
CO2	Students will be able to identify the issues involved in it.
CO3	Students to understand Methods used in Mobile Computing

Unit I : WIRELESS COOMUNICATION

[8 Hours]

Introduction to Wireless communication, Objectives, The cellular Concept, System design & fundamentals, Frequency reuse, Channel Assignment & handoff strategies, Adjacent Channel interference, cell splitting, Sectoring.

Unit II : INTRODUCTION TO GSM

[6 Hours]

Architecture, Radio Subsystem, Channel types, GSM frames structure, SDMA-FDMA-TDMA-CDMA, Cellular wireless networks, Wireless LAN-IEEE 802.11 standards: Architecture, services, MANET: Wifi & Wimax,

Unit III : MOBILE NETWORKS

[6 Hours]

Sensor Networks, Peer to Peer networks, mobile routing protocols:- DSR, AODV, reactive routing- Location Aided routing- Mobility models- Entity based-group mobility-Random ways point mobility model.

Unit IV : MOBILE NETWORKS LAYER

[8 Hours]

Mobile IP, dynamic host, configuration protocols, Adhoc Networks. MOBILE TRANSPORT LAYER: Traditional TCP, Indirect TCP, Snooping TCP, Mobile -TCP, Transaction oriented TCP.

Unit V : MOBILE NETWORKS

[8 Hours]

Issues & challenges- Security issues, Authentication in Mobile application, privacy issues, power management, Energy awareness computing, Mobile IP& Adhoc Network -VOIP application.

Text Books :-

1. Theodore S. Rappaport- Wireless Communication Principals & Practice Prentice Hall
2. Mobile Communication-Jochen Schiller, Adison Wesley, 2000
3. Wireless Communication & Networks-W. Stallings, Prentice Hall.

Reference Books :-

1. Lee. W. C.Y- Mobile Communication Engineering, Theory & Application ,II Edition
2. Principles of wireless Networks- Pahlavank & krishnamurthy. P
3. Mobile & wireless Networks-Black U.D.
4. Mobile Computing Technology, Applications and Service Creation Asoke K Talukder, Roopa R Yavagal, 2009, TATA McGraw HiLL









Subject : **Emerging Technology Lab-II**

Subject Code : **BTCME606P**

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
2 Hrs (Practical)	1	25	25	50

Practical List

Emerging Technology Lab-II is based on the following tools.

AI	Natural Language Toolkit (NLTK),SpaCy, PyTorch-NLP, Natural,Retext, TextBlob
DS	KNIME, Spark, Neo4J, MongoDB, Hive, Storm,
IoT	Devicehub, Zetta, Node-RED, Flutter, M2MLabs Mainspring









Subject : Mini Project & Internship

Subject Code : BTCME607P

Load	Credits	College Assessment Marks	University Evaluation	Total Marks
2 Hrs (Practical)	1	25	25	50

Course Objective:

1	To develop an understanding of applications in real life
2	To develop research skills of students
3	To help the students in exploring career opportunities in their areas of interest
4	To give an insight into the overall functioning of the organizations where students visited
5	To develop Industry Institute Interaction
6	To provide means to immerse students in actual supervised professional experiences.

Constraints:

1. Students shall work in groups of 4-5 each and work on small application or research based / Industry Oriented real time problems.
2. Local Mentor and Industry mentor shall work in coordination.
3. Industry visit should be planned to explore students about real time problems.
4. Students shall work on providing solutions to identified problems.
5. Detailed reports are expected to be submitted at the end using Standard Technical Writing Tool.
6. Evaluation should be done based on feedback of Local and Industry Mentor.

Expected Outcome:

1. Real Time Problem Identification
2. Requirement analysis and identification of relevant data sources
3. Literature survey / Industrial survey
4. Overall Project development as per the phases of SDLC



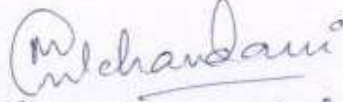
RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FOUR YEAR BACHELOR OF TECHNOLOGY (B.Tech) DEGREE COURSE
SEMESTER: V (C.B.C.S.)
BRANCH: COMPUTER SCIENCE AND ENGINEERING

Fifth Semester:-

S. N.	Subject	Teaching Scheme			Evaluation Scheme			Credits	Category
		L	T	P	CA	UE	Total		
1	Artificial Intelligence	3	1	-	30	70	100	4	PCC-CS
2	Artificial Intelligence-Lab	-	-	2	25	25	50	1	PCC-CS
3	Design & Analysis of Algorithms	3	1	-	30	70	100	4	PCC-CS
4	Design & Analysis of Algorithms -Lab	-	-	2	25	25	50	1	PCC-CS
	Software Engineering & Project Management	3	-	-	30	70	100	3	PCC-CS
5	Elective-I	3	-	-	30	70	100	3	PEC-CS
6	Effective Technical Communication	2	-	-	15	35	50	2	HSMC
7	Profesional Skills Lab I			2	25	25	50	1	ESC
8	Yoga and Meditation (Audit Course)	2	-	-	50	-	-	Audit	MC
	Total	16	02	06			600	19	

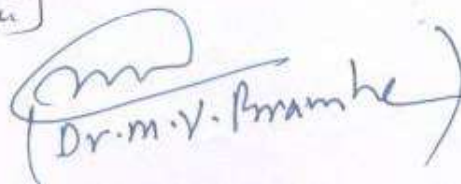
Elective-I: 1. TCP/IP 2. Design Patterns 3. Data Warehousing and Mining


 [Mrs. B. P. Dharekar]


 [Mrs. Mona Mulchandani]


 Dr. S. V. Sonekar


 Dr. S. V. Sonekar
 chairman


 (Dr. M. V. Pramhe)